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Impact of Pharmacist Preparation of Four-Factor Prothrombin Complex Concentrate at the Bedside in Patients with Life-Threatening Hemorrhage in the Emergency Department

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Abstract

Purpose: One option recommended by expert guidelines for prompt reversal of anticoagulation-induced hemorrhage is four-factor prothrombin complex concentrate (4F-PCC). Pharmacist presence has been shown to reduce order entry to administration and door-to-needle (DTN) times. However, how pharmacist preparation of 4F-PCC at the bedside in the Emergency Department (ED) can affect times to administration has not been thoroughly studied. The purpose of this study was to assess if pharmacist presence along with preparation of 4F-PCC at the bedside could improve administration times. **Methods:** This retrospective cohort study included anticoagulated patients requiring emergent reversal with 4F-PCC in the ED from January 2019 to April 2020 (pre-group) to March 2022 to March 2023 (post-group). Patients in the post-intervention group who had 4F-PCC prepared at the bedside were compared to a historical group without bedside 4F-PCC preparation. The primary outcome was time from 4F-PCC order entry to administration. **Results:** Of 193 patients evaluated, 99 (51.3%) were included (N = 41 pre-group; N = 58 post-group). There was a significant 11 minute difference in median time from order entry to administration favoring the post-group (31 vs 20 minutes, $P < .001$). The subset of patients with intracranial hemorrhage (ICH) in the post-group also had a significantly shorter order to administration time of 14 minutes (31 vs 17 minutes, $P < .001$). There was no difference in overall DTN times, in-hospital mortality, or length of stay (LOS) between groups, including in the subset of patients with ICH. **Conclusion:** 4F-PCC preparation at the bedside by the ED pharmacist significantly reduced 4F-PCC order entry to administration time. However, this intervention did not reduce overall DTN times.

Keywords
anticoagulation reversal, emergency department, time to treatment, pharmaceutical services, hemorrhage

Introduction

Vitamin-K antagonists (VKA) such as warfarin and direct oral anticoagulants (DOACs) such as apixaban, rivaroxaban, and dabigatran are commonly prescribed to treat and prevent embolic events. A common adverse effect of anticoagulation with these agents is hemorrhage, with a particular concern for intracranial hemorrhage (ICH). One option for prompt reversal of anticoagulation in patients who are actively hemorrhaging or for those requiring emergent procedural or surgical intervention is four-factor prothrombin complex concentrate (4F-PCC).[1-4] Four-factor prothrombin complex concentrate is approved for emergent reversal of VKAs in patients with acute major hemorrhage or need for emergent surgery or invasive procedures.[5] It is also commonly used off-label for the emergent reversal of DOACs in institutions that do not carry andexanet alfa.[1,2,4,6]

Despite the lack of literature describing optimal timing of anticoagulation reversal, it has been shown that faster reversal can improve patient outcomes, especially in patients with ICH. In an observational multicenter study, patients with ICH who had their international normalized ratio (INR) reversed to less than 1.3 with a systolic blood pressure less than 160 mmHg within 4 hours after admission had significantly fewer rates of hemorrhage expansion and in-hospital mortality.[7] In another trial, Hanger et al[8] showed that earlier reversal

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of VKA-associated ICH was associated with a trend to better survival, even after controlling for ICH score. The American Heart Association (AHA) Get With the Guidelines (GWTG)-stroke registry tracks performance measures that include the percentage of anticoagulated patients with life-threatening ICH who are reversed within 90 minutes of arrival.[9] Using this registry's database, in a retrospective cohort study including over 5000 patients with VKA and DOAC-associated ICH with known times of reversal, Sheth et al[10] found that an admission to administration ("door-to-needle"; DTN) time of 60 minutes or less was associated with decreased mortality and discharge to hospice.

However, 4F-PCC administration is often delayed due to operational and institutional barriers within the hospital and emergency department (ED). Various strategies for reducing delays in reversal have been studied and implemented. These include expanding 4F-PCC approval privileges to a larger group of physicians, utilizing point-of-care (POC) INR devices for faster bedside results, optimizing 4F-PCC storage locations, and implementing pharmacist-led practices.[11-16]

As reported by Masic et al[11] by incorporating these strategies along with the addition of ED pharmacists at the bedside, 4F-PCC order entry to administration times were reduced by a median of 9 minutes (34.5 vs 43.5 minutes, $P=.02$) and DTN times by a median of 140 minutes (66.5 vs 206.5 minutes, $P<.01$). However, this trial did not incorporate preparation of 4F-PCC at the bedside, and this intervention on times to administration has not been thoroughly studied in this setting. Benefits of pharmacist preparation at the bedside were reported for other time-sensitive medications, such as thrombolytics for the treatment of acute ischemic stroke (AIS) and pulmonary embolism, showing significantly shorter order to administration and DTN times with improvements in patient-centered outcomes.[17-21]

The purpose of this study was to assess the impact of pharmacist presence along with preparation of 4F-PCC at the bedside on order entry to administration times in patients with life-threatening hemorrhage (including ICH) or those requiring an emergent procedure or surgery.

Materials and Methods

This was a single-center, retrospective cohort study comparing 4F-PCC administration times in independent groups of patients seen before (pre-implementation) and after (post-implementation) implementation of a new protocol in which the pharmacist prepared 4F-PCC at the bedside in the ED.

Patients meet criteria for 4F-PCC use if they have a life-threatening hemorrhage, defined as acute and significant bleeding that causes hemodynamic compromise or high likelihood of causing death, disability, or the need for emergent operative or procedural intervention. Patients must also be anticoagulated with VKA (with a POC or laboratory INR of ≥2 or ≥1.4 with an ICH) or a DOAC (apixaban, rivaroxaban, edoxaban), and the condition must be survivable. If the hemorrhage is a central nervous system (CNS) or spine bleed without hemodynamic compromise, neurosurgery or the neurocritical care physician is contacted by the ED physician for reversal recommendations and the approval of 4F-PCC. Any patient with a hemorrhage causing hemodynamic compromise, including those with a CNS or spine bleed, automatically qualify for 4F-PCC, and no consulting service is contacted for approval. Our protocol recommends 4F-PCC at the FDA-approved variable-dosing strategy based on body weight and INR in patients taking VKA and the highest (50 units/kg) dose to reverse DOACs excluding dabigatran, which has its own reversal agent (idarucizumab) available at our institution.

Historically at our institution, after ED admission and the return of imaging results and laboratory values (specifically POC or laboratory INR or anti-Xa level) confirming anticoagulation-related hemorrhage, physicians enter a specific anticoagulation reversal orderset in the electronic medical record (EMR). Once an order for 4F-PCC is entered, since the exact number of factor IX units varies between vials, the ED pharmacist contacts the sterile IV pharmacy to determine the exact dose the patient will receive, adjusting the ordered dose to reflect nearest vial size as described by institutional policy. The order is then verified by the ED pharmacist, prepared in the sterile IV pharmacy located on the opposite side of the hospital, and hand-delivered to the ED nurse to administer via medication pump.

To analyze our process and establish baseline times to 4F-PCC administration, we reviewed the EMRs of consecutive patients admitted between January 2019 and April 2020; this served as our pre-implementation group. Implementation of a new protocol involving pharmacist preparation of 4F-PCC at the bedside was delayed due to the need to create and approve the new protocol and complete full education of all staff. Therefore, the finalized protocol was implemented in March 2022. This new protocol was identical to the process used for the pre-implementation group in terms of criteria for use, approval criteria, and order entry. However, with the new protocol, 4F-PCC was not made in a sterile IV pharmacy, rather, the ED pharmacist obtained 4F-PCC vials (which were moved to a closer location near the ED), adjusted the order to reflect nearest vial size, verified the order, and prepared the medication at the bedside. To ensure timely administration, the pharmacist also spiked and primed the tubing, delivered the medication to the nurse, and helped the nurse program the medication pump. If the ED pharmacist was unable to prepare the dose at the bedside for any reason (ie, Involved in other clinical tasks), they still had the option of verifying the dose to the sterile IV pharmacy as per the previous process. After a full year of implementing the new protocol (March 2022-March 2023), EMRs of consecutive patients were reviewed and analyzed as they compared to our pre-implementation group. As a sensitivity analysis, the results of the subset of patients with ICH were analyzed separately.

Included patients were anticoagulated, 18 years of age or older, and admitted to the ED requiring emergent reversal with 4F-PCC for either life-threatening hemorrhage or the need for an emergent procedure or surgical intervention. Patients were excluded if they were less than 18 years old, had completed neuroimaging confirming ICH at an outside institution, or received 4F-PCC at an outside hospital or location other than our institution's ED. Patients were also excluded if they were initially admitted to the ED for any reason other than known hemorrhage, even if they eventually received 4F-PCC. These patients required lengthy work-up and discovery of hemorrhage or need for emergent surgery or procedure occurred well after ED admission, potentially skewing time data. The research protocol was approved by our Institutional Review Board prior to data collection and informed consent was waived.

Our institution is an inner-city community safety net hospital serving as a Level 1 adult and pediatric trauma center and a comprehensive stroke center with 534 beds, 72 ED beds, and over 97 000 ED visits in 2023. Our ED is staffed daily with a clinical pharmacist from 7:00 AM to 1:00 AM then next day and covered by a separate, centrally located pharmacist between the overnight hours of 1:00 AM to 7:00 AM. This centrally located overnight pharmacy team was also trained on mixing 4F-PCC at the bedside. Emergency Medicine clinical pharmacy services were already in place prior to the development of this protocol, and no changes to the pharmacy-staffing model were necessary.

Endpoints

The primary outcome of interest was difference in 4F-PCC order to administration time between the pre- and post-implementation groups in a sample that included patients with ICH and without ICH. Order to administration time was defined as the length of time from order entry by the ED physician to the recorded start time of the 4F-PCC infusion in minutes. Secondary outcomes included ED admission to 4F-PCC administration time (DTN) and computed tomography (CT) confirmation of ICH (defined as the time the radiologist reported the results to the ED physician or finalized the read in the EMR) to 4F-PCC administration time in minutes for patients presenting with ICH. We also analyzed the proportion of patients achieving institution and GWTG-stroke DTN goals, hospital length of stay (LOS) in days, and in-hospital mortality rate.

Data Collection

Baseline characteristics collected include age, sex assigned at birth, race, weight, body mass index (BMI), and anticoagulation prescribed. We also noted the indication for anticoagulation reversal and location of hemorrhage. Additionally, we calculated the actual and weight-based (units/kg) dose of 4F-PCC given. For the post-implementation group, we collected the number of vials necessary to prepare the dose and we tallied the number of times each individual pharmacist prepared 4F-PCC at the bedside.

We collected the time stamps from the EMR using Epic (Verona, WI), which included times for ED admission, CT scan confirmation of hemorrhage (for the ICH population only), 4F-PCC order entry, order verification, and start of 4F-PCC administration. These data points were used to determine the total time from order entry to administration, the total time from CT scan confirmation of hemorrhage to administration for our ICH population, and the total DTN time. All study data were collected and managed using Microsoft Excel (Microsoft Corporation, Redmond, WA).

Statistical Analysis

Patient attributes, anticoagulation regimen, bleed location, 4F-PCC dose, time interval endpoints to administration, in-hospital mortality, and length of stay are described separately in the pre- and post-implementation groups using mean and standard deviation or median and inter-quartile range for continuous variables and count and percentages for categorical variables. Differences in patient attributes by pre- versus post-implementation group were tested with independent sample *t*-tests, Pearson's Chi-square test, and Fisher's exact test (for expected cell counts less than 5). All time interval endpoints and length of stay differences in pre- versus post-implementation groups were tested with the Wilcoxon rank sum test due to skewed distributions. Time from order to administration by pre- versus post-implementation group was also analyzed separately within type of anticoagulant (VKA vs DOAC). All analyses were repeated for the subset of 57 with ICH.

Among post-implementation patients only, time from order entry to administration was tested with the Wilcoxon rank sum test for pharmacists who reconstituted 4F-PCC 7 or more times versus fewer times, and for use of "low quantity" (2-3 vials) versus "high quantity" (4-5 vials). We chose 7 as the number of times a pharmacist was considered proficient at mixing 4F-PCC at the bedside and the differences in number of vials needed to prepare a dose to have a roughly even number between groups for analysis.

All tests were 2-sided and tests of group differences with $P < .05$ were considered statistically significant. Statistical analysis was conducted with SAS v9.4 (SAS Institute, Cary NC).

Data from a subset of the pre-implementation patients supplied assumptions used for the power analysis and the rationale for the amount of data to gather in the post-implementation period. In this sample of 39 patients the time from order entry to administration was 39 minutes (SD = 22). With 39 patients per group (pre- and post-implementation), a desired minimum detectable difference in groups of 15 minutes, and analysis of group differences with an independent samples *t*-test, this analysis would have power of 0.85

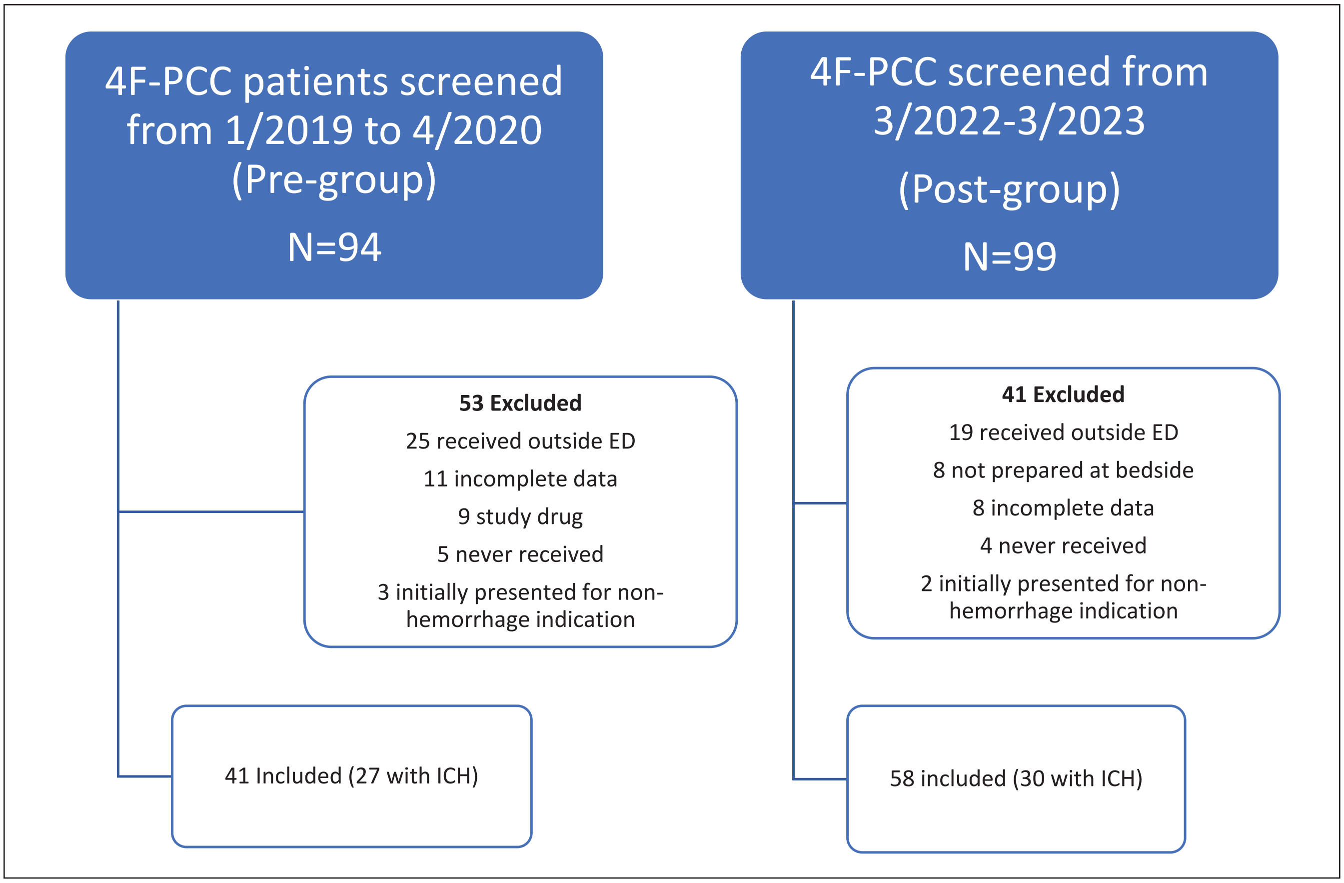

Figure 1. Flowchart of patients included.
Note. 4F-PCC = four-factor prothrombin complex concentrate; ED = emergency Department; ICH = intracranial hemorrhage.

(alpha = 0.05, 2-sided test). However, data collection continued for post-implementation patients beyond N = 39 to increase power for tests conducted of secondary outcomes.

Results

Of the 94 patients who received 4F-PCC in the pre- group, 41 (43.6%) were included in the study. The most common reasons for exclusion were patients receiving 4F-PCC in a location outside of the ED (47.2%, N = 25) and incomplete data (20.8%, N = 11). Of the 99 patients who received 4F-PCC in the post- group, 58 (58.6%) patients were included in the study. The most common reasons for exclusion were patients receiving 4F-PCC in a location outside of the ED (46.3%, N = 19) and either 4F-PCC was not made at the bedside or there was incomplete data (19.5%; N = 8 for both; Figure 1). Three patients in the pre-group and 2 patients in the post-group were excluded for reasons described above as they initially presented for non-hemorrhage indications. Nine patients in the pre-group were excluded from analysis as they were part of a separate 4F-PCC study that was ongoing at the same time. Eight patients in the post-group had 4F-PCC prepared in the sterile IV pharmacy instead of at the bedside as the ED pharmacist was involved in other clinical tasks.

The pre- and post-implementation groups had similar baseline characteristics. Notably, patients in the post- group were more likely to be anticoagulated with a DOAC as opposed to VKA than patients in the pre-group (17.1% vs 62.1%, $P < .001$). Both groups had similar hemorrhage locations. A large majority received 4F-PCC for a life-threatening hemorrhage, with emergent surgery or procedure (excluding neurosurgery for ICH) being the indication for reversal in 12.2% and 8.6% of patients in the pre- and post-groups, respectively. Only 1 patient in the pre-group and 2 in the post-group were not on anticoagulation and received 4F-PCC outside of protocol for an ICH (Table 1).

For our primary outcome, we found a statistically significant 11 minute difference in median time from order entry to administration favoring the post-group (31 vs 20 minutes, $P < .001$). Within the VKA group median time from order entry to administration favored the post group (32 vs 19 minutes, $P < .001$). Within the DOAC group median time from order entry to administration directionally favored the post group (31 vs 20 minutes, $P = .06$). There were no differences in DTN times between the groups, nor were there any

Table 1. Baseline Characteristics.

Characteristic	Pre-implementation N = 41	Post-implementation N = 58	*P*
Age, yrs, mean (SD)	79.2 (12.4)	77.3 (11.9)	.44
Male, N (%)	20 (48.8)	32 (55.2)	.53
White, N (%)	39 (95.1)	56 (96.6)	.99
Weight, kg, mean (SD)	80.3 (20.8)	85.7 (23.0)	.23
BMI, kg/m^2, mean (SD)	28.4 (6.4)	29.4 (7.1)	.47
Prescribed, N (%)			
Warfarin	34 (82.9)	22 (37.9)	<.001
Apixaban	3 (7.3)	26 (44.8)	
Rivaroxaban	3 (7.3)	8 (13.8)	
None	1 (2.4)	2 (3.5)	
Warfarin	34 (82.9)	22 (37.9)	<.001
Any DOAC	7 (17.1)	36 (62.1)	
Bleed location, N (%)			
Intracranial	27 (65.9)	30 (51.7)	.37
Gastrointestinal	5 (12.2)	11 (19.0)	
Intra-abdominal	2 (4.9)	4 (6.9)	
Genitourinary	0 (0)	2 (3.5)	
Thoracic	1 (2.4)	0 (0)	
Other	1 (2.4)	6 (10.3)	
Reversal for urgent surgery/procedure, N (%)	5 (12.2)	5 (8.6)	.56

Note. P-values from independent samples *t*-test for continuous variables, Pearson's Chi-square test or Fisher's exact test for categorical variables. SD = standard deviation; BMI = body mass index; DOAC = direct oral anticoagulant.

Table 2. Characteristics of 4F-PCC Administration.

Characteristic	Pre-implementation N = 41	Post-implementation N = 58	*P*
4F-PCC dose (units), median (IQR)	2200 (1737-2712)	3168 (2275-4268)	<.001
4F-PCC dose (units/kg), median (IQR)	27.0 (25.0-31.3)	46.1 (27.1-51.7)	<.001
4F-PCC order entry to admin time (min), median (IQR)	31 (25-44)	20 (15-28)	<.001
ED presentation to administration (DTN; min), median (IQR)	111 (79-172)	93.5 (54-149)	.22
In-hospital mortality, N (%)	9 (22.0)	14 (24.1)	.80
Length of stay, day, median (IQR)	6 (4-7)	6 (4-7)	.66

Note. P-values from Wilcoxon rank sum test and Pearson's Chi-square test. 4F-PCC = four-factor prothrombin complex concentrate; IQR = interquartile range; ED = emergency department; DTN = door-to-needle.

differences in in-hospital mortality rate or hospital LOS (Table 2).

When analyzing results for the ICH population separately, baseline characteristics were also well-matched between groups. As in the full cohort, patients in the post-group were more likely to be anticoagulated with a DOAC as opposed to VKA than patients in the pre-group (14.8% vs 66.7%, $P < .001$; Table 3).

Patients with ICH in the post-group had a significantly shorter order entry to administration time of 14 minutes (31 vs 17 minutes, $P < .001$). Within the VKA group median time from order entry to administration favored the post group (29 vs 16 minutes, $P = .001$). Within the DOAC group median time from order entry to administration directionally favored the post group (36 vs 18 minutes, $P = .06$). There was also a statistically significant 25.5 minute reduction in CT scan confirmation of ICH to administration time in the post-group (61 vs 35.5 minutes, $P = .04$). In addition, a higher percentage of patients in the post-group achieved institution goal times of CT scan confirmation of ICH to administration of 45 minutes or less (37% vs 63.3%, $P = .047$; Table 4).

A comparison of order entry to administration time in the post-implementation group (N = 58) by pharmacists who had reconstituted 4F-PCC at the bedside 7 or more times (N = 34) versus those with 6 or fewer times (N = 24) showed no difference (median = 19.4 vs 20.0, $P = .80$, Wilcoxon rank sum test). We also analyzed if the number of 4F-PCC vials requiring reconstitution affected time from order entry to

Table 3. Baseline Characteristics of Patients with ICH.

Characteristic	Pre-implementation N = 27	Post-implementation N = 30	*P*
Age, yrs, mean (SD)	80.9 (11.3)	78.2 (11.9)	.37
Male, N (%)	13 (48.2)	18 (60.0)	.37
White, N (%)	26 (96.3)	29 (96.7)	.99
Weight, kg, mean (SD)	74.0 (13.4)	82.3 (19.7)	.07
BMI, kg/m^2, mean (SD)	26.6 (3.7)	28.2 (6.1)	.24
Prescribed, N (%)			
Warfarin	23 (85.2)	10 (33.3)	<.001
Apixaban	2 (7.4)	14 (46.7)	
Rivaroxaban	1 (3.7)	4 (13.3)	
None	1 (3.7)	2 (6.7)	
Warfarin	23 (85.2)	10 (33.3)	<.001
Any DOAC	4 (14.8)	20 (66.7)	
Location of ICH, N (%)			
Multiple	9 (33.3)	8 (26.7)	.95
Intraparenchymal	7 (25.9)	10 (33.3)	
Subdural	8 (29.6)	7 (23.3)	
Subarachnoid	3 (11.1)	4 (13.3)	
Intraventricular	0 (0)	1 (3.3)	

Note. P-values from independent samples *t*-test for continuous variables, Pearson's Chi-square test or Fisher's exact test for categorical variables. ICH = intracranial hemorrhage; SD = standard deviation; BMI = body mass index; DOAC = direct oral anticoagulant.

Table 4. Characteristics of 4F-PCC Administration in Patients With ICH.

Characteristic	Pre-implementation N = 27	Post-implementation N = 30	*P*
4F-PCC dose (units), median (IQR)	2100 (1650-2272)	3168 (2635-4268)	<.001
4F-PCC dose (units/kg), median (IQR)	26.2 (25.0-30.3)	45.8 (27.1-52.1)	.003
4F-PCC order entry to admin time (min), median (IQR)	31 (24-41)	17 (12-26)	<.001
Admission to CT scan read (min), median (IQR)	36 (23-62)	45 (24-82)	.64
CT scan read to administration (min), median (IQR)	61 (35-85)	35.5 (20-58)	.04
CT scan read to administration within 45 min, N (%)	10 (37.0)	19 (63.3)	.047
ED presentation to administration (DTN; min), median (IQR)	86 (77-137)	83 (51, 128)	.30
DTN times within 90 min, N(%)	15 (55.6)	16 (53.3)	.87
DTN times within 60 min, N (%)	5 (18.5)	10 (33.3)	.20
In-hospital mortality, N (%)	8 (29.6)	11 (36.7)	.57
Length of stay, days, median (IQR)	6 (4-11)	6 (4-16)	.85

Note. P-values from Wilcoxon rank sum test and Pearson's Chi-square test. 4-FPCC = four-factor prothrombin complex concentrate; ICH = intracranial hemorrhage; IQR = interquartile range; CT = computed tomography; ED = emergency department; DTN = door-to-needle.

administration in the post-group. When stratified as "low quantity" (2-3 vials) or "high quantity" (4-5 vials), there was no difference between groups.

Discussion

To the authors' knowledge, routine preparation of 4F-PCC at the bedside by a pharmacist is not a common practice in the ED setting. As previously reported, pharmacist presence alone can reduce order entry to administration and DTN times for 4F-PCC.[11,12] Our study adds evidence that implementing a protocol for pharmacist presence along with preparation of 4F-PCC at the bedside can further reduce order to administration time. These are very promising results, and other centers may see an even greater benefit than ours owing to our already efficient pre-implementation process. For example, in one small Japanese study, Imanaka et al[22] showed that including pharmacists who also prepared 4F-PCC at the bedside led to a 39 minute reduction in order entry to administration time (21 vs 60 minutes, $P = .02$) in patients on VKA therapy requiring emergent reversal.

We specifically chose order entry to administration time as our primary outcome because this is the part of the care continuum of a hemorrhaging patient pharmacists can influence most. We did find preparing 4F-PCC at the bedside can improve those times as well as CT scan confirmed ICH to administration times. However, in centers that already involve pharmacist presence in the ED, there may be a ceiling effect for how much preparing 4F-PCC at the bedside can improve overall DTN times. For example, in the Imanaka et al trial,[22] although a significant reduction in order entry to administration time was seen, there was no effect on overall DTN times (103 vs 111 minutes, $P=.4$). We report a similar median of 111 minutes in the pre-group and 93.5 minutes in the post-group in DTN times in the full sample. In the ICH subgroup, DTN times were relatively improved but also not different between the pre- and post-groups (86 vs 83 minutes, $P=.3$).

Expert guidelines do not provide optimal reversal time goals for any hemorrhaging patient, including those with an ICH. The GWTG- stroke group tracks centers on proportion of patients with ICH meeting DTN time of less than 90 minutes, with a plan to match the AIS guidelines of a DTN goal less than 60 minutes in the future.[9] We were able to meet the 90 minute goal only half of the time in both the pre- and post-groups, with a non-statistically significant improvement in the patients achieving the 60 minute goal in the post-group (18.5% vs 33.3%, $P=.20$).

Achieving stringent DTN goals of less than 90, 60, 45, or even 30 minutes to mimic AIS guidance will necessitate optimization of the entire care continuum of patients with ICH, likely involving the combined efforts of prehospital and inpatient systems. Institutions must implement quality improvement initiatives that strive to improve DTN times by focusing on factors in addition to pharmacist presence and preparation of 4F-PCC at the bedside. One rate-limiting factor is reducing time to imaging. Using the AIS recommendation of admission to CT imaging time of 25 minutes will be a point of focus for institutions attempting to improve care for their ICH population. We showed a median time of 36 and 45 minutes in the pre- and post-groups, respectively, in time from ED admission to imaging, whereas others have reported much longer times of up to 186 minutes.[8]

In the full sample and among the subgroup of patients with ICH, patients in the post-group were significantly more likely to be prescribed a DOAC than those in the pre-group. Our institution guideline recommends the maximum 50 unit/kg dose for the reversal of DOACs. This explains why patients in the post-group received higher absolute and weight-based doses of 4F-PCC. These higher doses naturally necessitated a greater number of vials needing reconstitution, which was nevertheless overcome by preparing the medication at the bedside. This was corroborated by our analysis that showed that even with the increasing number of vials needing reconstitution, there was no difference in time from order entry to administration. In addition, a second analysis found that pharmacist experience with the protocol did not influence time from order entry to administration. We noted that our regular staff ED pharmacists who prepared 4F-PCC at the bedside had similar times to administration as those who do not staff the ED regularly.

More patients in the pre-group were prescribed VKA, which could prolong time to administration while awaiting INR results. However, reduction in time to administration comparing the post to pre-group was similar in the VKA and DOAC groups for the whole sample and for the ICH sample as well, suggesting that the reduction in administration times occurred regardless of type of anticoagulant the patient was taking. This may be due in part to the fact that our institution utilizes POC INR testing initially while awaiting confirmation from laboratory INR. Likewise, in patients presumably taking DOACs there is also often a need to confirm use and compliance by ordering and awaiting anti-Xa levels to result, offsetting the INR result delay in patients taking VKAs. Guidelines recommend the use of anti-Xa levels (calibrated with low-molecular-weight heparin or unfractionated heparin) to exclude clinically important levels of DOACs in institutions, like ours, that do not have anti-Xa assays that are calibrated to the specific DOAC of interest.[1]

There is much complexity in the presentation, workup, diagnosis, and treatment of a hemorrhaging patient. This often includes detailed histories, medication reconciliation, laboratory measurements, imaging, discussions with consulting services, goals of care discussions, and reversal product ordering and administration. The addition of clinical pharmacists and preparation of 4F-PCC at the bedside is a crucial, although singular optimization of the care of the hemorrhaging patient. Larger prospective trials will be necessary to determine how to further improve administration times and ascertain optimal DTN times for 4F-PCC administration.

Limitations

This single-center retrospective chart review has several limitations, including a small sample size. In addition, administration time data was pulled from the EMR, with the understanding that documentation of critical emergent interventions may not be entered in real time during life-threatening situations. Similarly, CT scan results in patients with ICH may have been discussed verbally prior to documenting the final read in the EMR, but time stamps from the EMR were used to control for differences between cases. Given the retrospective nature, we were unable to determine how many patients in either group required neurosurgery or neurocritical care provider approval of 4F-PCC, nor how long it took to obtain this approval. However, given there was no difference in bleed location (including number of patients with ICH) between the groups, we presume that the number of patients needing 4F-PCC approval, and the time it takes to obtain approval, would be similar between groups.

Finally, these patients were the first to receive 4F-PCC under the new process from ED pharmacy staff with variable prior experience with 4F-PCC preparation. Although we found no difference in pharmacist experience and times to administration, prolonged use of this protocol may show that with more experience, some users may show faster administration times.

Conclusion

In this retrospective analysis of real-world use of 4F-PCC in a large level 1 trauma and comprehensive stroke center, 4F-PCC preparation at the bedside by the ED pharmacist significantly reduced 4F-PCC order entry to administration time, CT-confirmed ICH to administration time, and led to a higher proportion of patients meeting the institution goal of CT-confirmed ICH to administration time of less than or equal to 45 minutes. There were no differences in DTN times, in-hospital mortality rates, or LOS.

Key Points

- Pharmacist preparation of medications at the bedside leads to faster administration for time-sensitive medications.
- There is a lack of data for how much bedside preparation of four-factor prothrombin complex concentrate (4F-PCC) can affect times to administration in patients with life-threatening hemorrhage.
- Pharmacist preparation of 4F-PCC at the bedside can significantly reduce order entry-to-administration times, also importantly in the subset of patients with intracranial hemorrhage, but does not affect overall door-to-needle times.

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Author Contributions

Adis Keric - Conceptualization, Methodology, Investigation, Supervision, Data Curation, Writing- Original Draft, Writing-Reviewing & Editing, Visualization, Project Administration.
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Casey Woster, MD -Conceptualization, Methodology, Writing-Review & Editing.
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Data Availability

The entire deidentified dataset, dictionary, and code for this investigation are available upon request by contacting Adis Keric, PharmD at the email address listed above.

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Ethical Considerations

This study received ethical approval from the HealthPartners IRB on January 22, 2022. This is an IRB-approved retrospective study, all patient information was de-identified and patient consent was not required. Patient data will not be shared with third parties.

Consent to Participate

Informed consent was waived by the HealthPartners IRB.

Consent for Publication

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